Ji et al. 2020 — Cutaneous Squamous Cell Carcinoma

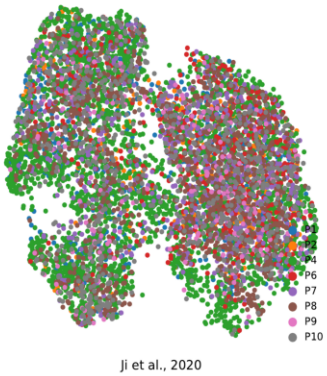
10x scRNA-seq · 8/10 patients retained · 7,085 malignant developmental cells (10,531 pre-scope)

BRANCHING — Gast/pSom hub → Gast arm + pSom arm · Crest terminal on pSom arm · hub r = −0.47

GO / NO-GO GATES

Gate 1 — mixing & continuity	**GO**	θ=10 integrates 8 patients into one connected manifold; no per-sample islands.
Gate 2 — embryonic module geography	**GO**	Gast/pSom hub diverges into opposed Gast & pSom arms; Crest terminal on pSom arm.
Gate 3 — hub cross-repression (branching)	**GO**	Mixed-hub Gast↔pSom mutual r = −0.47 (Puram −0.49); null 0.01±0.09, p<0.003.

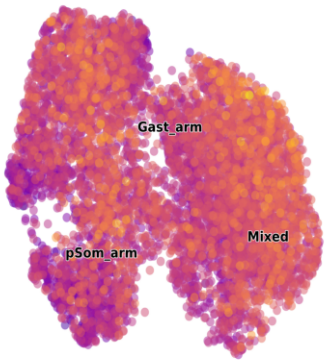
Samples (integration)



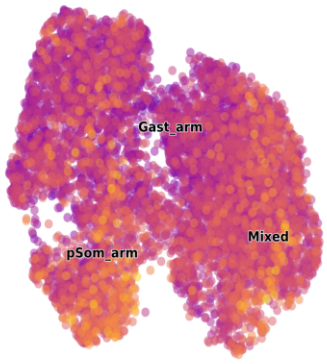
Manifold + trajectory



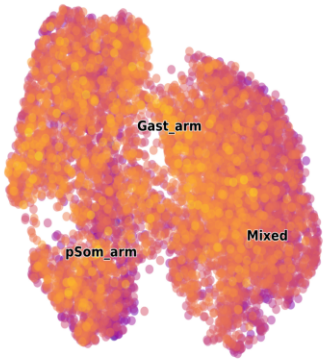
Gast signature



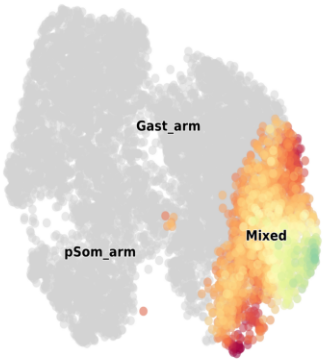
pSom signature



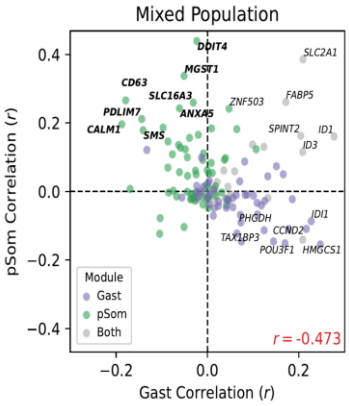
Crest signature



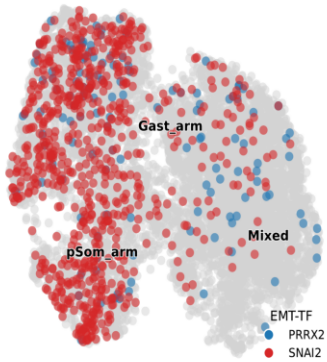
Hub module correlation



Hub cross-repression (r)



EMT-TF expression



PARAMETER CHOICES & RATIONALE

Threshold ≥100 cells/sample — clean 6.70× count gap (40→268); keeps 8/10 patients (10,531 cells).

Genes 1,443 overdispersed (scFates find_overdispersed) · PCA 50 PC.

Integration Harmony on sample, θ=10 (10x precedent). Branch survives strong integration ⇒ biology, not a per-patient archipelago.

Cycling scope 3,446 cells in 5 high-cycling (>50%) Leiden lobes removed; 7,085-cell developmental fork re-embedded (Filbin protocol).

Embedding kNN n_pcs=20 / n_neighbors=30 · UMAP spread=1.3.

Trajectory scFates PPT tree Nodes=60 λ=3000 σ=0.5 → 4 tips / 2 forks (hub-and-spoke Y).

Hub fork-Leiden(res 0.4) cluster of max joint (min-z) Gast+pSom co-expression = "Mixed" (1,513 cells).

Root Gast/pSom-high, Crest-low tip (root_score 0.99); the Crest-high tip is a terminus, not the root.

EMT-TFs sparse 10x detection (3% floor): PRRX2 (Gast arm) ↔ SNAI2/SLUG (pSom arm terminus, neural-crest EMT-TF).

Verdict Embryonic branching pattern REPRODUCES — Gast/pSom hub, two cross-repressing arms, Crest terminal on pSom arm (r = −0.47 ≈ Puram −0.49).